RESULTS

Optimal Context Mapping

A memory beside the AI model that gives it the right context first, and gets cheaper every time a question comes back.

Devieswar Kancheti · 2026

THE PROBLEM

AI research is paid for again and again

Every round costs more

An AI agent re-reads everything it has found each time it asks the model again.

Repeats cost full price

The same fact asked in new words means the whole search starts over.

Every word is billed

Companies pay for every word the model reads and every request it makes.

Optimal context mapping hands the model the right context first, so it searches less and pays less.

WHAT IT ACHIEVED

Against Claude with search tools

A stream of 120 research questions over collections of 1M, 3M and 9M tokens.

3–5×

fewer model requests

1.1–1.3×

fewer tokens per right answer

Same

accuracy

118 vs 120 · 117 vs 117 · 117 vs 116 right, of 120

WHEN QUESTIONS COME BACK

Repeat research is where it shines

The same facts, asked again in completely new words:

1.7–1.8× **fewer tokens**

4.5–9.1× **fewer requests**

It gets cheaper the more it is used.

By collection size

Size	All questions tokens · requests	Repeats tokens · requests
1M	1.34× · 3.5×	1.83× · 4.9×
3M	1.10× · 5.0×	1.68× · 9.1×
9M	1.11× · 3.0×	1.79× · 4.5×

How many times fewer than Claude with search tools, per right answer.

HOW IT WAS TESTED

A fair race

Fresh agents

New Claude Opus 5.5 agents in every run, with the same instructions except the memory.

Blind judges

Every answer graded by judges who did not know which agent gave it.

Rules fixed first

Each test plan was written and locked before it ran, so nothing could be tuned after.

Held-out questions

60 questions never seen while building it, over public text.

Counted, not estimated

Every token and request taken from the agents' own logs.

Losses reported

Where it did not win, the paper says so.

WHERE IT HELPS

Best wherever knowledge is asked for again

Helps most

Teams researching the same documents: the same facts, asked again in new words.

Support and knowledge bases: many people asking the same thing in different ways.

Compliance and legal re-checks: the same policies looked up again and again.

AI products billed per request: 3–5× fewer requests means faster answers and lower bills.

Honest limit

Brand-new questions are not cheaper in tokens the first time, though they still take 1.5–2.7× fewer requests.

The saving grows as questions repeat.

OPTIMAL CONTEXT MAPPING

The same right answers, with 3–5× fewer requests, and cheaper every time a question returns.

Full paper: "Optimal Context Mapping", D. Kancheti, 2026